

QUIZ 1

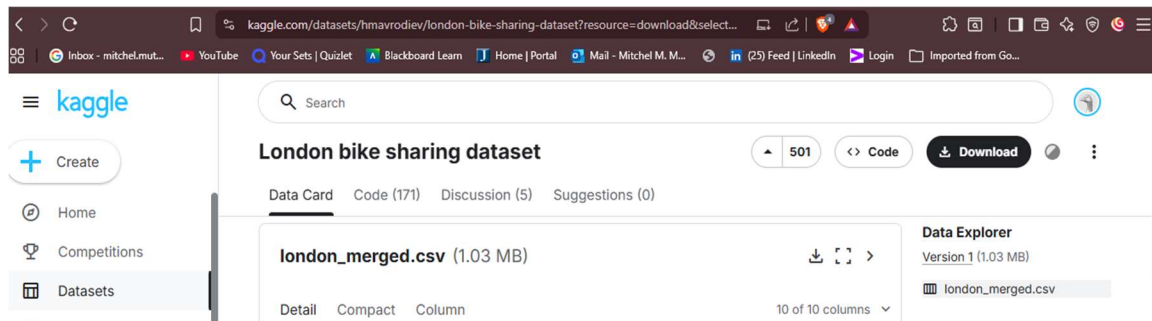
Name: Mitchel Makena

ID: 413

Data Source: London Bike

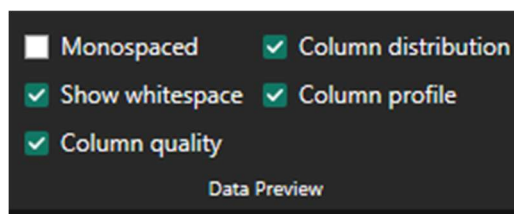
Link: https://www.kaggle.com/datasets/hmavrodiev/london-bike-sharing-dataset?resource=download&select=london_merged.csv

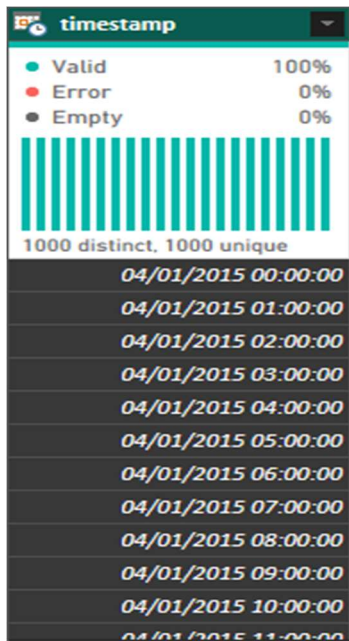
Dataset



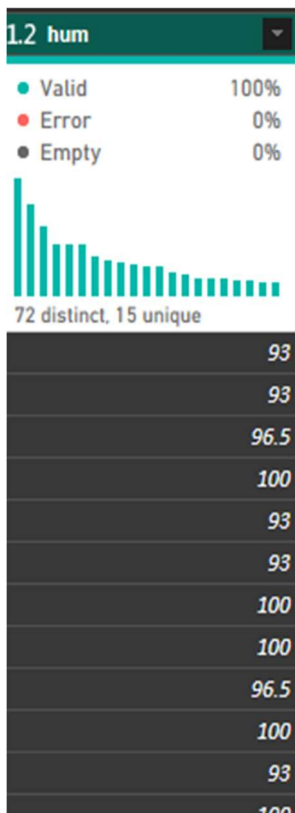
Section A

Q1



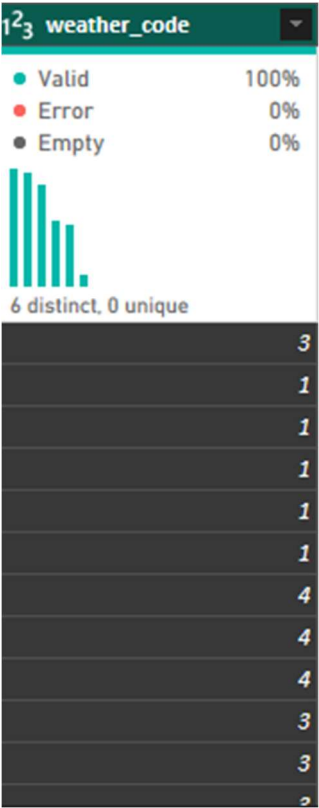


High cardinality (too many unique values) makes it hard to aggregate. Fix: Split Time and Date.



Percentage stored as a whole number (e.g., 80 instead of 0.8). Fix: Divide by 100.

Numeric codes (1, 2, 3) are not descriptive. Fix: Merge with a look up.

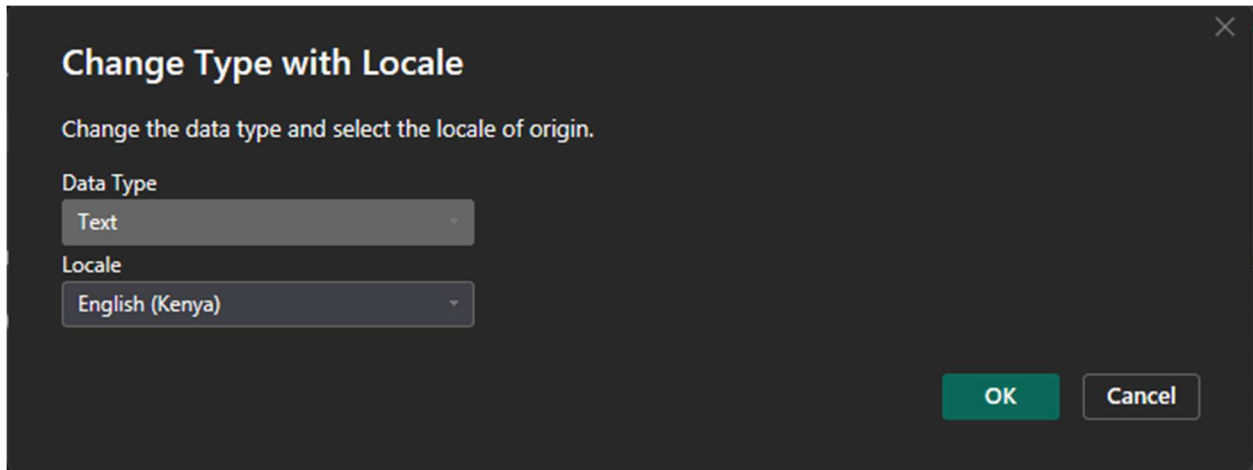




Non-descriptive names. Fix : Rename.

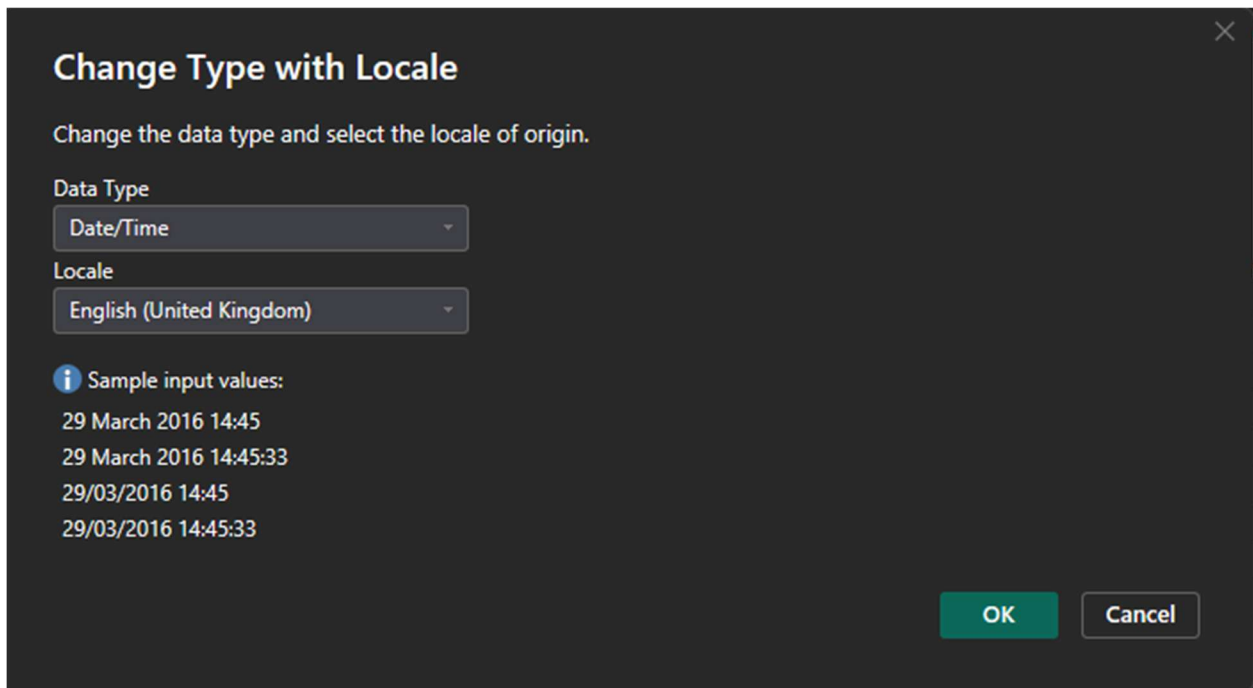
Q2

Before :



A dark-themed dialog box titled "Change Type with Locale" with a close button (X) in the top right corner. Below the title is the instruction "Change the data type and select the locale of origin." There are two dropdown menus: "Data Type" with "Text" selected and "Locale" with "English (Kenya)" selected. At the bottom right are "OK" and "Cancel" buttons.

After :



A dark-themed dialog box titled "Change Type with Locale" with a close button (X) in the top right corner. Below the title is the instruction "Change the data type and select the locale of origin." There are two dropdown menus: "Data Type" with "Date/Time" selected and "Locale" with "English (United Kingdom)" selected. Below these is an information icon (i) followed by the text "Sample input values:" and a list of four sample values: "29 March 2016 14:45", "29 March 2016 14:45:33", "29/03/2016 14:45", and "29/03/2016 14:45:33". At the bottom right are "OK" and "Cancel" buttons.

Q3

×

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

Space

Split at

☐ Left-most delimiter

☐ Right-most delimiter

☒ Each occurrence of the delimiter

▸ Advanced options

Quote Character

"

☐ Split using special characters

Insert special character ▾

OK

Cancel

Date	Time_hour
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
42 distinct, 0 unique	24 distinct, 0 unique
04/01/2015	00:00:00
04/01/2015	01:00:00
04/01/2015	02:00:00
04/01/2015	03:00:00
04/01/2015	04:00:00
04/01/2015	05:00:00
04/01/2015	06:00:00
04/01/2015	07:00:00
04/01/2015	08:00:00
04/01/2015	09:00:00
04/01/2015	10:00:00
04/01/2015	11:00:00
04/01/2015	12:00:00
04/01/2015	13:00:00

Trim and Clean :

Date	Time_hour	Bike_Count
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
42 distinct, 0 unique	24 distinct, 0 unique	727 distinct, 535
04/01/2015	00:00:00	
04/01/2015	01:00:00	
04/01/2015	02:00:00	
04/01/2015	03:00:00	
04/01/2015	04:00:00	
04/01/2015	05:00:00	
04/01/2015	06:00:00	
04/01/2015	07:00:00	
04/01/2015	08:00:00	
04/01/2015	09:00:00	
04/01/2015	10:00:00	
04/01/2015	11:00:00	
04/01/2015	12:00:00	
04/01/2015	13:00:00	

PROPERTIES

Name
london_merged

All Properties

APPLIED STEPS

- Source
- Promoted Hea...
- Changed Type
- Divided Column
- Filtered Rows
- Renamed Colu...
- Changed Type ...
- Changed Type1
- Split Column b...
- Changed Type2
- Renamed Colu...
- Trimmed Text
- Cleaned Text**

Q4

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Demand_Category

	Column Name	Operator	Value		Output	
If	Bike_Count	is greater than or...	2000	Then	High Demand	...
Else If	Bike_Count	is greater than or...	800	Then	Medium Demand	

Add Clause

Else
Low Demand

OK Cancel

Custom Column

Add a column that is computed from the other columns.

New column name
Temp_Gap

Custom column formula
= [Actual_Temp] - [Temp]

Available columns

- Date
- Time_hour
- Bike_Count
- Actual_Temp
- Temp
- hum
- wind_speed

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Q5

Merge

Select a table and matching columns to create a merged table.

london_merged

Date	Time_hour	Bike_Count	Actual_Temp	Temp	hum	wind_speed	weather_code	is_holi
04/01/2015	00:00:00	182	3	2	0.93	6	3	
04/01/2015	01:00:00	138	3	2.5	0.93	5	1	
04/01/2015	02:00:00	134	2.5	2.5	0.965	0	1	
04/01/2015	03:00:00	72	2	2	1	0	1	

Weather

Code	Description
1	Clear
2	Broken Clouds
3	Rainy
4	Cloudy
7	Windy

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 17414 of 17414 rows from the first table.

OK Cancel

Search Columns to Expand

☒ Expand ☐ Aggregate

☒ (Select All Columns)

☐ Code

☒ Description

☐ Use original column name as prefix

OK Cancel

SECTION B

Q6



Star Schema Identification :

- **Fact Table:** London_Fact

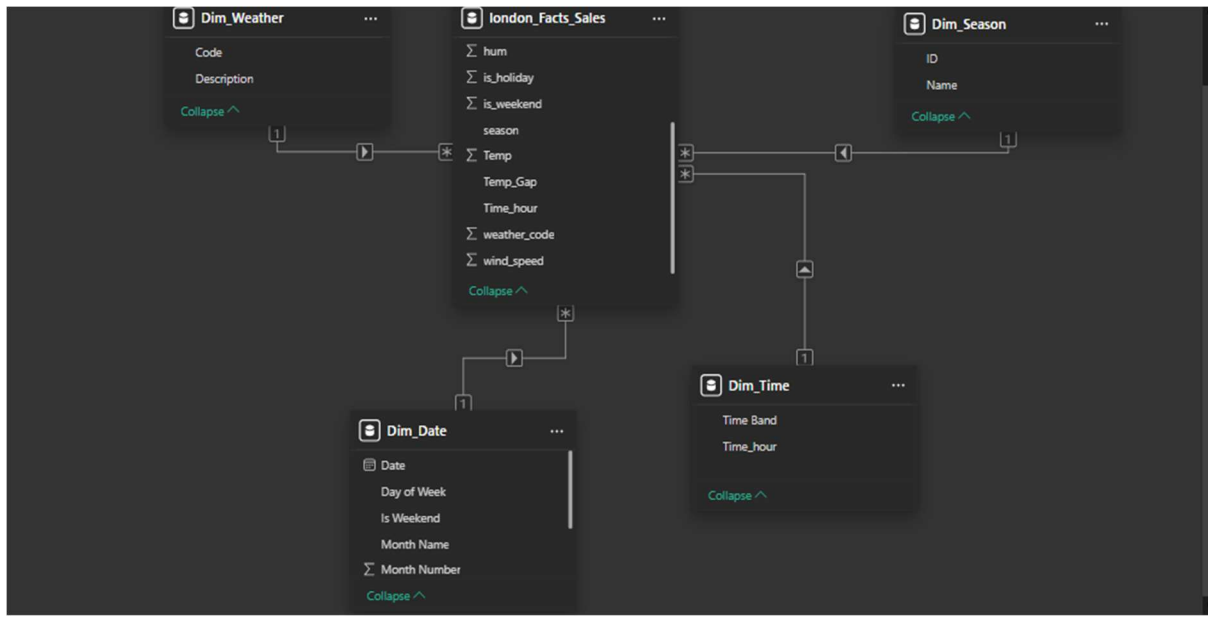
Justification: The quantitative observations (measures) of bike rentals and environmental data are included in this table. It links to every lookup table and has the greatest grain.

- **Dimension Table:** Dim_Weather (Mapping Table)

Justification: The descriptive qualities (such as "Clear" and "Rainy") in this table provide the fact table's numerical codes context.

- **Dimension Table:** Dim_Date (Calendar Table)
- **Dimension Table:** Dim_Season (Season Table)
- **Dimension Table:** Dim_Hour (Hours Table)

Q7



Relationship 1: Dim_Weather to Fact[weather_code]

- Cardinality: One-to-Many
- Cross Filter Direction: Single
- Status: Active
- Why correct: Each numeric weather code appears once in the lookup table (Dimension) but can be recorded multiple times (Many) in the rental logs (Fact).

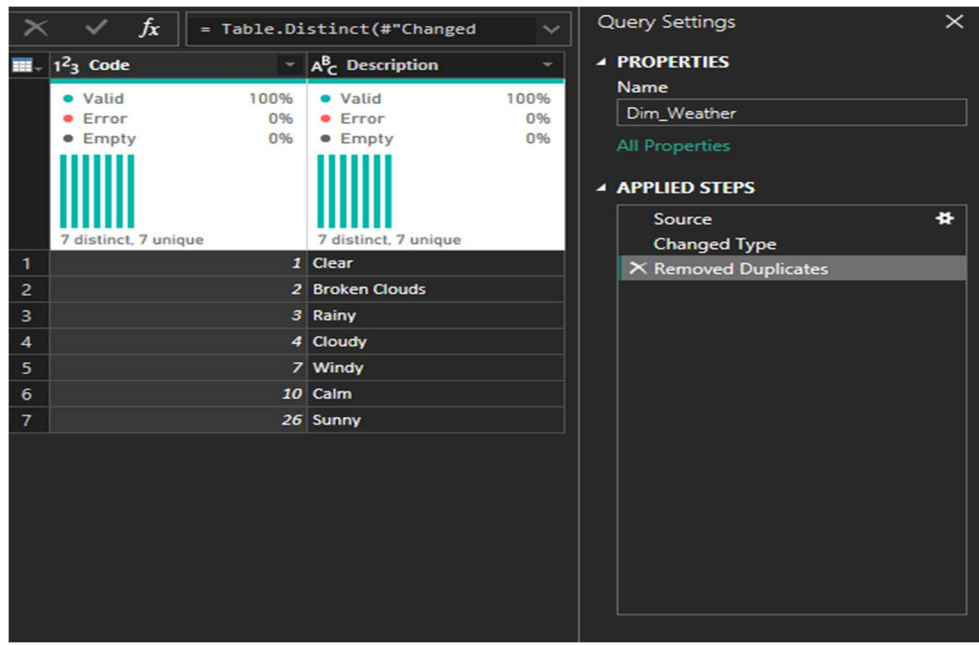
Relationship 2: Dim_Time [Hour] to Fact [Time_Hour]

- Cardinality: One-to-Many
- Cross Filter Direction: Single
- Status: Active
- Why correct: This allows us to slice thousands of rental records by a specific hour (e.g., 8:00 AM) to analyze peak commute patterns.

Identify One modelling mistake:

- **Mistake:** Connecting Dim_Season directly to Dim_Weather, snow only happens in winter.
- **Impact:** This creates a Snowflake Schema rather than a Star Schema. While logically true, it increases complexity and can slow down report performance in Power BI compared to having both tables connect directly to the Fact table.

Q8



Verification Method: In Power Query, I selected the Code column in Dim_Weather, went to the View tab, and checked Column Distribution. It confirmed "100% Unique."

Action Taken: Even though it was unique, I applied the "Remove Duplicates" step on the primary key column to ensure that if the source file is updated with redundant rows in the future, the model will not break with a "Many-to-Many" error.

Q9

- **Decision:** Yes, a data table must exist.
- **Reasons:**
 - i) **Reporting Flexibility:** It allows us to group data by attributes not found in the raw data, such as "Fiscal Year," "Month Short Name," or "Day of the Week."
 - ii) Allows for **Year-over-year comparisons** of Bike Rentals.
 - iii) **Time Intelligence:** Without a dedicated Date table, Power BI cannot reliably perform DAX functions like SAMEPERIODLASTYEAR or TOTALYTD because the timestamp in the London data is hourly, not a continuous daily sequence.
- **Columns to include:** Date, Month Name, Year.

Bonus: Small Dictionary

Table	Column	Meaning	Type	Key/Attribute
London_Fact	Bike_Count	Count of new bike shares	Whole Number	Measure
London_Fact	weather_code	Numeric ID for weather	Whole Number	Foreign Key
Dim_Date	Date	Unique calendar date	Date	Primary Key
Dim_Date	Is Weekend	Flag for Saturday/Sunday	Text	Attribute
Dim_Weather	Description	Clear, Rainy, Snow, etc.	Text	Attribute
Dim_Time	Hour	24-hour clock (0-23)	Whole Number	Primary Key
Dim_Time	Time Band	Period (e.g., Rush Hour)	Text	Attribute
Dim_Season	Season Name	Spring, Summer, etc.	Text	Attribute

VISUALIZATION

